Homework-03

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Set Up

```r
# load in packages
library(tidyverse)
library(janitor)
library(readxl)
library(here)
library(ggeffects)
```

```r
# store kelp data as an object called kelp
kelp <- read_csv(here("data","temp-kelp.csv"))
# store egg data as an object called egg_data
egg_data <- read_csv(here("data", "personal-data-project - Sheet1.csv"))
```

Problems

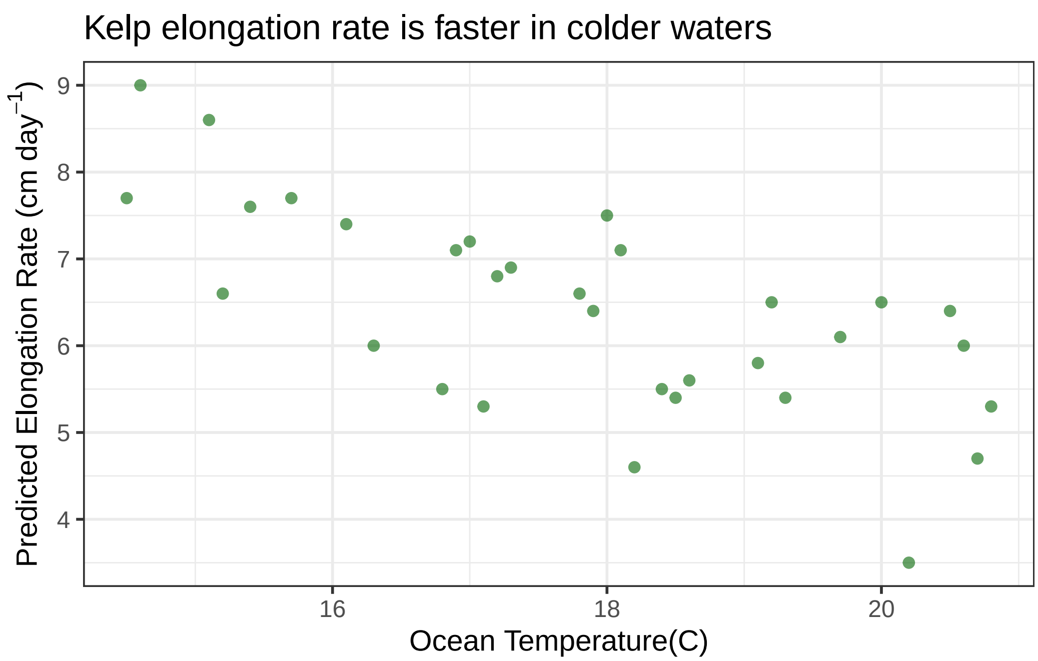
Problem 1: Giant Kelp Fronds

a. An appropriate test

To determine the strength of the relationship between temperature and elongation rate, you could do a Pearson's correlation and a Spearman rank correlation. A Pearson's correlation is parametric and is best for understanding the strength of the correlation between variables when there is a linear relationship between continuous variables. A Spearman's rank correlation is non-parametric and useful for understanding the strength of the correlation between variables that show a monotomic relationship between variables.

b. Create a visualization

```
# base layer: ggplot
ggplot(data = kelp, # use kelp data frame
       aes(x = temp_c, # x-axis: temperature (C)
           y = kelp_elong)) + # y-axis: kelp elongation rate (cm/day)
  # 1st layer: show raw data points
  geom_point(color = "darkgreen", # points = dark green
             alpha = 0.6) + # points = slightly transparent
  # label and and y axes
  labs(x = "Ocean Temperature(C)",
       y = expression(paste("Predicted Elongation Rate (cm ", day^-1,")")),
       # add title
       title = "Kelp elongation rate is faster in colder waters") +
  # change theme
  theme_bw() +
  # remove legend
  theme(legend.position = "none")
```



c. Check and run test

Create Linear Model

```
# create new object called kelp model to store linear model
# function: lm
# formula: dependent ~ independent, data = data frame
kelp_model <- lm(kelp_elong ~ temp_c, data = kelp)
```

```
# show summary of linear model
summary(kelp_model)
```

```
Call:
lm(formula = kelp_elong ~ temp_c, data = kelp)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.8643 -0.5017  0.1790  0.5659  1.2177 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 14.08645    1.49219   9.440 1.72e-10 ***
temp_c      -0.43179    0.08321  -5.189 1.37e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom
Multiple R-squared:  0.473,	Adjusted R-squared:  0.4554 
F-statistic: 26.93 on 1 and 30 DF,  p-value: 1.366e-05
```

```
# show in a table
flextable::as_flextable(kelp_model)
```

	Estimate	Standard Error	t value	Pr(>|t|)	
(Intercept)	14.086	1.492	9.440	0.0000	***
temp_c	-0.432	0.083	-5.189	0.0000	***

	Estimate	Standard Error	t value	Pr(>|t|)

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

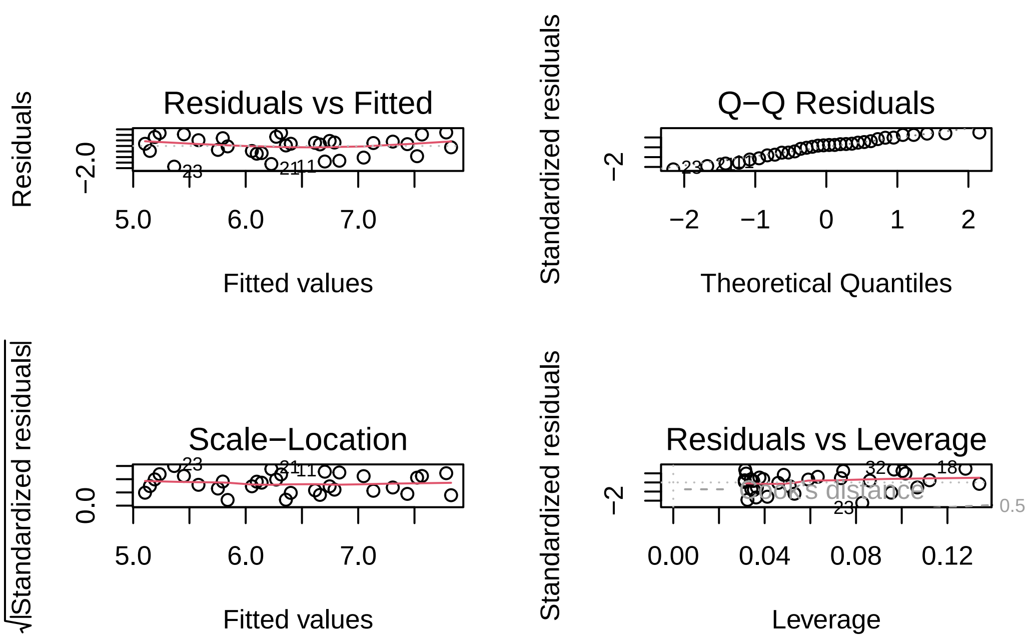
Residual standard error: 0.8665 on 30 degrees of freedom

Multiple R-squared: 0.473, Adjusted R-squared: 0.4554

F-statistic: 26.93 on 30 and 1 DF, p-value: 0.0000

Check Assumptions

```r
# create 2x2 grid to show diagnostic plot
par(mfrow = c(2,2))
# show diagnostic plots
plot(kelp_model)
```



The QQ residuals plot shows the data points following the line. This suggests the distribution of the residuals is normal. The residuals vs. fitted and scale locations plots don't seem to show a pattern. The data seems to be randomly and evenly distributed which indicates that the residuals are homoscedastic. The residuals vs leverage plot does not show points outside of the dashed lines which indcates that there are no outliers influencing the model.

Run Test

```r
# run pearson's correlation test
# # formula: (data set$independent, data set$dependent)
kelp_corp <- cor.test(kelp$temp_c, kelp$kelp_elong, method = "pearson")
# show output
show(kelp_corp)
```

```

	Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

d. Results communication

To evaluate the strength of the relationship between temperature and kelp frond elongation, I used Pearson's correlation (r). This test is a good choice because both ocean temperature and elongation rate are continuous variables, and the diagnostic checks show that the data satisfies the assumptions of normality and linearity. I suspect a linear relationship between ocean temperature and kelp frond elongation rate

I found a strong negative relationship between ocean temperature and giant kelp frond elongation rate (Pearson's r = -0.7, t(30) = -5.189, $p < 0.001$, $\alpha = 0.05$).

e.

I would tell them that ocean temperature and the growth rate of kelp are related and that kelp fronds in high temperature ocean conditions tend to have a slower growth rate. I would also tell them that more tests need to be done in order to fully describe the relationship between temperature and growth rate.

f. Double check your own work

```
# run spearman's rank test
kelp_cors <- cor.test(kelp$temp_c, kelp$kelp_elong, method = "spearman")
# show output
show(kelp_cors)
```

```

	Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6891684
```

The Spearman's rank correlation would have led me to make the same decision - that there is a relationship between ocean temperature and kelp frond elongation rate.

We found a strong negative relationship between ocean temperature and giant kelp frond elongation rate (Spearman ρ = -0.69, S = 9216.1, p $<$ 0.001, α = 0.05).

Spearman rank correlation is the non-parametric version of the Pearson's correlation. The Spearman test is better suited to variables that are monotonic and it converts the data in to ranks instead of comparing the raw data like Pearson's r does.

Problem 2: Personal Data

```
# create new object from egg_data
egg_data_clean <- egg_data |>
  # clean names
  clean_names() |>
  # remove any row where all values are NA
  remove_empty(which = "rows") |>
  # remove the empty (1st) column that was all NA
  select(-1) |>
  # change names from 1, 2, 3, etc.
  set_names("date", "number_eggs_eaten", "servings_yogurt_eaten",
```

```
  "breakfast_at_school", "servings_meat_eaten", "school_day") |>
  # remove the first row (old spreadsheet cloumn titles)
  slice(-1) |>
  # convert the columns from text to numbers
  mutate(number_eggs_eaten = as.numeric(number_eggs_eaten),
         servings_yogurt_eaten = as.numeric(servings_yogurt_eaten),
         servings_meat_eaten = as.numeric(servings_meat_eaten))
```

```
# base layer: ggplot
# use clean data set
# set x-axis to school day
# set y-axis to number of eggs eaten
# make the point color different based on school day
# make the fill color of the box plot different based on school day
ggplot(data = egg_data_clean,
       aes(x = school_day,
           y = number_eggs_eaten,
           color = school_day,
           fill = school_day)) +
  # 1st layer: geom_boxplot
  # make the boxplot fill more transparent
  # make the lines of the boxes black
  # hide the boxplot outlier
  geom_boxplot(alpha = 0.6,
               color = "black",
               outlier.shape = NA) +
  # 2nd layer: geom_jitter
  # make the individual dots visible but still within their category (0.1)
  # make the points transparent enough to see if they overlap
  geom_jitter(width = 0.1,
              alpha = 0.7) +
  # change boxplot color
  # change point color
  scale_fill_brewer(palette = "Accent") +
  scale_color_brewer(palette = "Accent") +
  # label x and y axes
  # add title
  labs(x = "School Day",
       y = "Number of Eggs Eaten",
       title = "Effect of Going to School on the Number of Eggs I Eat",
       subtitle = "Date of last collection: 2026-05-26") +
  # change the theme
```

```
  theme_light() +
      # center title and make text bold
theme(plot.title = element_text(hjust = 0.5,
                                face = "bold"),
      # center subtitle
      plot.subtitle = element_text(hjust = 0.5),
      legend.position = "none") # remove the legend
```

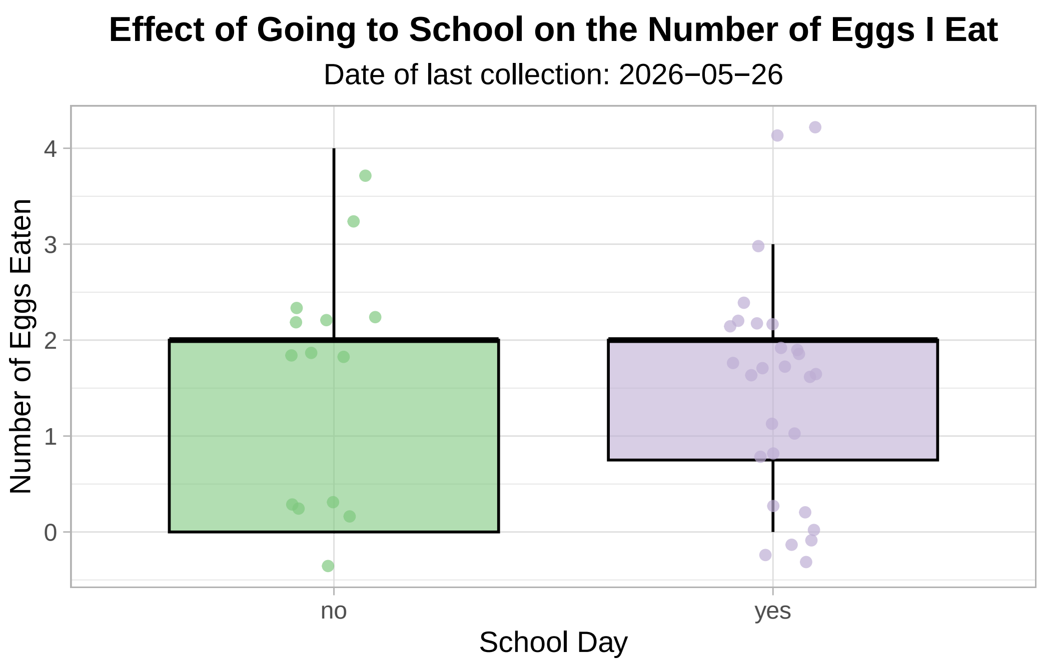


Figure 1. Distribution of the daily number of eggs consumed on days with and without school. The green represents the number of eggs eaten on non-school days. The purple represents the number of eggs eaten on school days. Individuals data points are shown as circles in their respective color. The tops of both boxes represent the median and 75th percentile and the bottoms limits of the boxes represent the 25th percentile. The whiskers indicate points 1.5 times beyond the interquartile range.

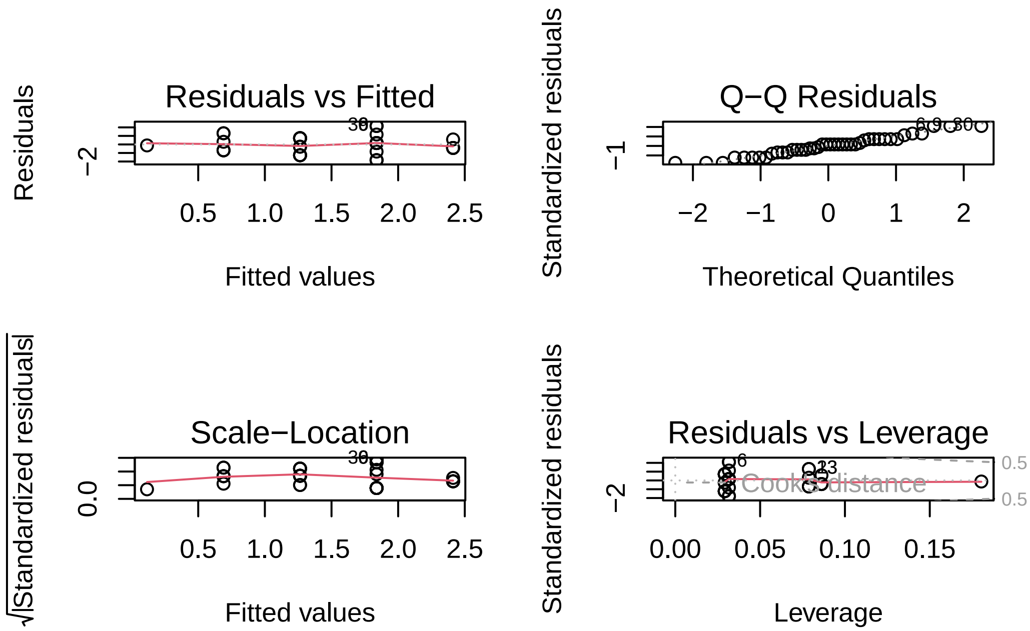
Create egg linear model

```
# create new object to store linear model
# formula: lm(dependent ~ independent, data = data frame)
egg_model <- lm(number_eggs_eaten ~ servings_meat_eaten,
                data = egg_data_clean)
# show summary of linear model
summary(egg_model)
```

```
Call:
lm(formula = number_eggs_eaten ~ servings_meat_eaten, data = egg_data_clean)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.8382 -0.6900  0.1618  0.7359  2.1618 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)           2.4123     0.3072   7.851 1.26e-09 ***
servings_meat_eaten  -0.5741     0.1689  -3.399  0.00154 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.047 on 40 degrees of freedom
Multiple R-squared:  0.2241,	Adjusted R-squared:  0.2047 
F-statistic: 11.55 on 1 and 40 DF,  p-value: 0.001543
```

Check Assumptions

```r
# create 2x2 grid to show diagnostic plot
par(mfrow = c(2,2))
# show diagnostic plots
plot(egg_model)
```



New Figure

```
# create new object to store predictions of the model
egg_predict <- ggpredict(egg_model, terms = "servings_meat_eaten")
# base layer: ggplot
ggplot(data = egg_predict, # use predictions data frame
       aes(x = x, # x-axis: x
           y = predicted # y-axis: predicted
)) +
  # 1st layer: geom_point to show raw data points
  geom_point(data = egg_data_clean,
             aes(x = servings_meat_eaten,
                 y = number_eggs_eaten), #
             color = "orange", # point color = dark green
             alpha = 0.5) + # make points slightly transparent
  # 2nd layer: ribbon to show 95% CI
  geom_ribbon(aes(ymin = conf.low, # lower bound of 95% CI
                  ymax = conf.high), # upper bound of 95% CI
              fill = "orange3",
              alpha = 0.3) + # make ribbon transparent
  # 3rd layer: line representing model predictions
  geom_line(color = "red2") +
  # relabel x and y axes and add title
```

```r
labs(x = "Servings of Meat Eaten",
     y = "Number of Eggs Eaten",
     title = "Increased meat consumption predicts
     decreased egg consumption",
     subtitle = "Date of last collection: 2026-05-26") +
# change theme
theme_bw() +
# center title and make text bold
theme(plot.title = element_text(hjust = 0.5,
                                face = "bold"),
      # center subtitle
      plot.subtitle = element_text(hjust = 0.5))
```

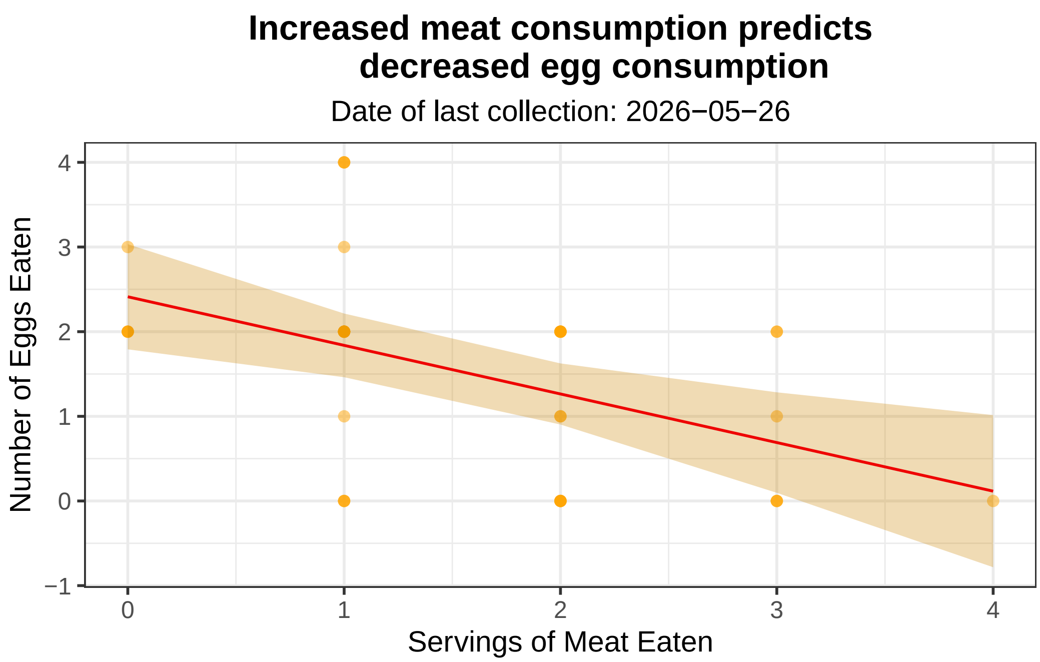


Figure 2. Relationship between daily meat consumption and the number of eggs eaten. As the number of daily meat servings increase, the number of eggs consumed decreases. Individual daily observations are represented by orange points. The red solid line indicated the ordinary least squares linear regression trend. The orange ribbon represents the 95% confidence interval.

Problem 4: Statistical critique

a. Revisit and summarize

The authors are using an ANOVA to assess the effect of clearing on floral resources available for pollinators and on pollinator communities. The ANOVA compares the variance within the cleared plots to the variance between the cleared and uncleared plots to assess whether or not there is a significant difference in floral resources and arthropods between the two treatments.

```r
knitr::include_graphics(here("data","pollinator-boxplot.png.png"))
```

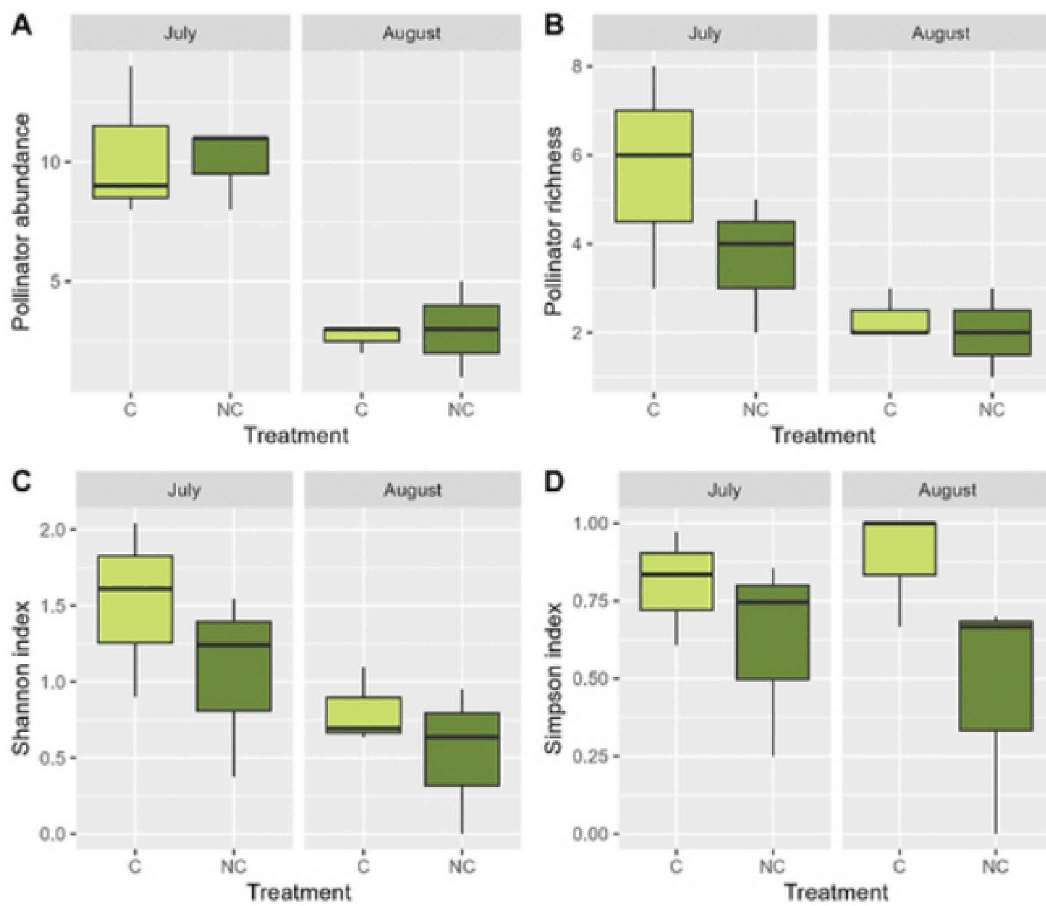


Figure 1: "Boxplots of A) pollinator abundance, B) pollinator richness, C) Shannon index and D) Simpson index for sampling-time (July, August) and treatment (C= cleared, NC= not cleared)."

b. Visual clarity

The figure represents the statistics in the figure reasonably clearly. It is split up into 4 sections. Each section compares a difference dependent variable to the treatment. Within each section there are two facets - one that contains data for July and one that contains data from August. The axes are logically positioned. On the x-axis is the treatment (cleared or non-cleared land) and on the y-axis is A) pollinator abundance, B) pollinator richness, C) Shannon index, D) Simpson index. Basically, the figure is comparing the pollinator presence and the treatments. The boxplots show the median and interquartile range but no underlying data is shown

c. Aesthetic clarity

I think the authors could have done a better job of minimizing visual clutter. There are 8 panels and 16 boxplots in the figure. They could have made the figure more concise and therefore the data to ink ration is not as low as it could be.

d. Recommendations

They should take off the gray background from the headers and the panel and instead make the text saying "July" or "August" bold.

They should take off the "Treatment" label from the top two plots and just keep the label on the very bottom of the page.

They should take out the vertical grid lines because there are only two treatments so all the grid lines aren't adding to the figure.